

Final Testing Report — FR-01: Account Registration

Assignment: HW02 — AI-First Domain Testing & Boundary Value Analysis

Feature ID: FR-01

Feature: Account Registration

Date: 2026-07-07

AI Tool: Antigravity (Claude Sonnet 4.6 Thinking)

SUT: EShop — <http://localhost:5173>

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1. Feature Overview

Item	Detail
Feature	Account Registration
Actor	Guest (unauthenticated user)
Entry Point	http://localhost:5173/register
API Endpoint	POST http://localhost:3000/api/register

Item	Detail
Expected Success Response	HTTP 200 — <code>{"message": "User registered successfully", "id": <n>}</code>

Inputs

Input	Variable	Required	Constraints (from UI)
Họ Tên	name	Yes	Non-empty
Email	email	Yes	Valid email format; must be unique
Mật khẩu	password	Yes	≥ 8 chars, uppercase, lowercase, digit, special char

Key Observations from DOM Inspection

- Email field uses `type="text"` — **no native browser email format validation**
- All fields have `maxlength = -1` — **no HTML-level length limits**
- All fields have `required = true` — browser blocks submission if blank

2. Testing Environment

Component	URL	Status
Frontend	http://localhost:5173	✅ Reachable
Backend	http://localhost:3000	✅ Reachable
Playwright	headless Chromium	✅ Working

Evidence: [ENV-register-page.png](#)

Đăng Ký Tài Khoản

Họ Tên

Email

Mật khẩu

Yêu cầu: Tối thiểu 8 ký tự, có chữ hoa, chữ thường, số và ký tự đặc biệt.

Đăng Ký

Đã có tài khoản? [Đăng nhập](#)

3. Domain Testing Summary

Methodology

Technique: **Equivalence Partitioning / Domain Testing**

Artifacts: DT-01 → DT-02 → DT-03 → DT-04 (each reviewed via REVIEW-01)

Partitions Identified

Variable	Valid Partitions	Invalid Partitions	Total
name	1	1	2
email	1	5	6
password	1	6	7
Total	3	12	15

Test Cases

TC ID	Scenario	Partitions Covered	Expected	Actual	Status
TC001	All valid inputs	NAME-V1, EMAIL-V1, PASS-V1	✔ Registered	✘ "Mật khẩu quá yếu!"	FAIL
TC002	Empty name	NAME-I1	✘ Rejected	✘ Browser tooltip	PASS
TC003	Empty email	EMAIL-I1	✘ Rejected	✘ Browser tooltip	PASS
TC004	Email no @	EMAIL-I2	✘ Rejected (email error)	✘ Password error shown	FAIL
TC005	Email no domain	EMAIL-I3	✘ Rejected (email error)	✘ Password error shown	FAIL
TC006	Email no local part	EMAIL-I4	✘ Rejected (email error)	✘ Password error shown	FAIL
TC007	Duplicate email	EMAIL-I5	✘ Rejected (duplicate)	✘ Password error shown	FAIL
TC008	Empty password	PASS-I1	✘ Rejected	✘ Browser required	PASS
TC009	Password < 8 chars	PASS-I2	✘ Rejected	✘ "Mật khẩu quá yếu!"	PASS
TC010	No uppercase	PASS-I3	✘ Rejected	✘ "Mật khẩu quá yếu!"	PASS
TC011	No lowercase	PASS-I4	✘ Rejected	✘ "Mật khẩu quá yếu!"	PASS
TC012	No digit	PASS-I5	✘ Rejected	✘ "Mật khẩu quá yếu!"	PASS
TC013	No special char	PASS-I6	✘ Rejected	✘ "Mật khẩu quá yếu!"	PASS

4. Boundary Value Analysis Summary

Applicable Variables

Variable	Boundaries	Reason Included/Excluded
name	None observable	✗ Skipped
email	None observable	✗ Skipped
password	Min = 8 chars (UI stated)	✓ Applied

BVA Test Cases

TC ID	Boundary	Value	Expected	Actual	Status
BVA-TC001	min – 1 (7 chars)	Pas1!Aa	✗ Rejected	✗ Rejected	PASS
BVA-TC002	min (8 chars)	Pas1!Aab	✓ Accepted	✗ Rejected	FAIL
BVA-TC003	min + 1 (9 chars)	Pas1!Aabc	✓ Accepted	✗ Rejected	FAIL

BVA Result: 1 PASS / 2 FAIL

Both failures caused by BUG-001 — boundary correctness cannot be independently confirmed.

5. Execution Results

Overall

Phase	Total TCs	PASS	FAIL
Domain Testing	13	8	5
BVA	3	1	2

Phase	Total TCs	PASS	FAIL
Total	16	9	7

Notable Observations

- 1. **API never reached** — all validation is client-side. BUG-001 blocks every submission.
- 2. **Generic error message** — "*Mật khẩu quá yếu!*" is shown for all failures including invalid emails.
- 3. **Email field is type="text"** — no native browser email validation.

Scripts

Script	Purpose
playwright/exec_fr01_dt.js	Domain Testing execution (13 TCs)
playwright/exec_fr01_bva.js	BVA execution (3 TCs)

6. Bug Reports

BUG-001 — Critical

Field	Value
ID	BUG-001
Title	Valid password Password1! wrongly rejected as too weak
Severity	● Critical
Impact	Completely blocks all new user registrations
Evidence	bugs/FR01/screenshots/TC001-after.png
File	bugs/FR01/BUG-001.md

Description: The client-side password validator rejects any password, including those that satisfy all 5 stated rules (length ≥ 8, uppercase, lowercase, digit, special character). The API is never called.

BUG-002 — High

Field	Value
ID	BUG-002
Title	No email format validation on registration form
Severity	 High
Impact	Users can register with malformed email addresses
Evidence	bugs/FR01/screenshots/TC004-after.png , TC005-after.png , TC006-after.png
File	bugs/FR01/BUG-002.md

Description: The email field uses `type="text"` . Invalid formats (`invalidemail` , `user@` , `@example.com`) are not rejected by the client. No custom JS validation is present.

BUG-003 — High (Blocked)

Field	Value
ID	BUG-003
Title	Duplicate email registration not detected
Severity	 High
Status	Blocked — requires BUG-001 fix to confirm
Evidence	bugs/FR01/screenshots/TC007-after.png
File	bugs/FR01/BUG-003.md

Description: When a previously-registered email is submitted again, no duplicate-detection error appears. Masked by BUG-001 — API never reached.

7. AI Gap Analysis

Full analysis in `tests/FR01/GAP-01-gap-analysis.md` .

Category	Result
Missing confirmed test cases	0
Hallucinated requirements	0
Incorrect expected results	0
Reasoning errors	0
Candidate gaps (pending human review)	3 (whitespace name, whitespace password, email leading space)

Overall AI quality: High. All rules were grounded in observable evidence. Assumptions were flagged explicitly. BUG-001 masking was correctly identified without assuming implementation details.

8. Conclusion

Feature Status: ✖ **BROKEN**

The Account Registration feature (FR-01) is currently **non-functional** due to BUG-001. No user can successfully register via the UI.

Recommended Priority

Priority	Action
1	Fix BUG-001 (password validator logic)
2	Fix BUG-002 (add email format validation)
3	Re-test TC001 and TC007 after BUG-001 fix
4	Re-test BVA-TC002 and BVA-TC003 to confirm boundary behaviour
5	Verify BUG-003 (duplicate email) after BUG-001 fix

Artifacts Index

Artifact	Path
ENV-01	tests/FR01/ENV-01.md
DT-01 Feature Understanding	tests/FR01/DT-01-feature-understanding.md
REVIEW-01 of DT-01	tests/FR01/REVIEW-01-of-DT-01.md
DT-02 Domain Identification	tests/FR01/DT-02-domain-identification.md
REVIEW-01 of DT-02	tests/FR01/REVIEW-01-of-DT-02.md
DT-03 Domain Partitioning	tests/FR01/DT-03-domain-partitioning.md
REVIEW-01 of DT-03	tests/FR01/REVIEW-01-of-DT-03.md
DT-04 Test Cases	tests/FR01/DT-04-test-cases.md
EXEC-01 (Domain Testing)	tests/FR01/execution.md
BVA-01 Boundary Analysis	tests/FR01/BVA-01-boundary-analysis.md
EXEC-01 (BVA)	tests/FR01/execution-bva.md
GAP-01 Gap Analysis	tests/FR01/GAP-01-gap-analysis.md
BUG-001	bugs/FR01/BUG-001.md
BUG-002	bugs/FR01/BUG-002.md
BUG-003	bugs/FR01/BUG-003.md